

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000094
Award Program	2018 Seeding Solutions: Addressing Challenges in Food and Agriculture Funding Opportunity
Project Title	Bridging the Genome-to-Phenome Breeding Gap for Water-Efficient Crop Yields (G2P Bridge)
Grant Start Date	01/01/2019
Grant End Date	12/31/2022
Total Grant Budget	\$2,012,321.35
Total FFAR Budget	\$999,578.90
Matching Funder(s)	\$1,012,744.32
Final Report Date	10/03/2023
Project Director/Principal Investigator Information	
Full Name	Geoffrey P Morris
Email	geoff.morris@colostate.edu
Phone	(970) 491-2102
Grantee Organization Information	
Organization Name	Colorado State University
Address	2002 Campus Delivery
City, State Zip	Fort Collins, CO 80523-2002
Tax ID	84-6000545
Authorized Signing Official (ASO) Information	

Name	Catherine Douras
ASO Title	Senior Research Administrator
Email	Catherine.douras@colostate.edu
Phone	970-491-2375

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
 - Manhattan KS - 1st Congressional District.
 - Lubbock TX - 19th Congressional District.
 - Parlier, CA - 21st Congressional District.
 - Fort Collins, CO - 2nd Congressional District
- 1.2. How many new jobs were created because of this grant funding?
1
- 1.3. How many jobs were maintained because of this grant funding?
3
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of

the project help to address important needs in U.S. food and agriculture systems.

Water scarcity is a longstanding challenge for U.S. agriculture and producers are looking for more diversified farming systems that provide a balance of water conservation and water productivity. Our project, using the genome-to-phenome (G2P) breeding approach integrated crop modeling, genomic prediction, and managed-stress experiments to develop improved water productivity for underutilized sorghum lines. Under this approach we characterized two genotype-by-management systems hypothesized to increase productivity under water scarcity: i) early planting of chilling tolerant hybrids, shifting the growing season to an earlier and more favorable environmental water demand, and ii) limited transpiration (LT), shifting the season to a late and improved synchrony with plant water demand and use. Our findings support the effectiveness of the G2P approach over conventional breeding for water-limited environments using sorghum as a case study and increased the understanding and implementation of the G2P approach in the field crop breeding community. In addition, via in silico simulations we demonstrate that introducing chilling tolerance in commercial hybrids allows for early planting and improved grain yields (6 - 11%). Currently KASP markers for chilling tolerance were developed and this technology is in process to be adopted by private seed companies. Additional simulations with the ILT trait, which has a similar soil water-saving mechanism to the stay-green trait, revealed a potential yield increase from 3 to 11%. These results suggest that the introgression of LT is more valuable in western regions of the sorghum belt, where farmers predominantly plant mid and short-season sorghums. The results of this project outline a path forward for producers looking to diversify their farming systems and adapt to changing environmental stressors that pose a risk to traditional crop systems.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

As previously stated, with the impending increase in water scarcity and limited access to water resources, producers will need to pivot their operations and consider alternative crops that tolerate water limited environments, and use

water resources more effectively, shifting to a more sustainable model moving forward. The results of our chilling tolerance trials indicate that sorghum may prove to be a preferred alternative for producers and has high potential to be readily adopted. If a chilling-tolerant sorghum line was made commercially available to producers, we could more readily move away from monoculture plantings and diversify the types of crops being grown, stagger planting and harvest dates to ease workload demands, increase grain yields and income on sorghum crops, mitigate risk and income loss due to drought, and conserve water resources.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

Based on our findings, and effective use of the G2P approach in sorghum breeding, our first recommendation is that crop breeding programs adopt the G2P approach when planning their experimental design and work through defining the temporal and spatial target population of environments (TPE) to identify traits and conditions that result in breeding experiments that more effectively lead to favorable phenotypes. This concept and systematic approach can lead to the discovery of novelty traits and accelerate genetic gains.

Our second recommendation is to support the development of chilling tolerance and LT in breeding programs. Sorghum is less relevant than major crops (i.e., maize and soybean), a reason why it is planted late. Introducing early chilling tolerance in sorghum can provide farmers with an alternative to avoid terminal drought.

- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

Early soil moisture in spring can secure seed germination for



sorghum planted in early spring. Locations with high snow cover can benefit from introducing chilling tolerance when planted early. For instance, a breeder/collaborator pointed out that Nevada has this characteristic. Thus, chilling tolerance could increase the farmer's incomes in this state.

- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

No

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

GOAL1: Use crop models to characterize the target population of environments (TPE) and predict scenarios under which chilling tolerance and stay green loci provide water-limited yield.

Objective: Calibration and model testing for biomass accumulation, flowering time, and commercial sorghum yield under traditional management practices.

Objective: Hypothesize trade-offs of early-planted chilling tolerant sorghum and dense-planted stay-green sorghum hybrid-based.

GOAL 2: Complete development of genomic prediction resources, including loci, markers, and germplasm.

Objective: Genotype and generate genomic predictions for genome-wide polygenic effects and known major effect loci (height, maturity, chilling tolerance, and stay-green loci).

Objective: Map water-limited yield loci using training population genotyping data and phenotyping data.

Objective: Develop KASP markers for novel water-limited yield loci.

Objective: Advance genomic prediction population to C1-F7 generation and



C2-S2 generation

GOAL3: Collect a third year of managed-stress phenotyping data for the crop model calibration panel and a first year of data for the updated (C1) training population.

Objective: Collect data on environment (soil texture, soil water, weather) and crop performance (yield, phenology, biomass) for commercial and experimental hybrids (N = 40) from CA (managed-stress) field sites.

Objective: Phenotype testcross hybrid updated (C1) training population (N = 200) under managed stress (CA) and in sorghum TPE (TX and KS).

GOAL4: Train early-career researchers (project postdocs and workforce development participants) on technical aspects of G2P approach components and identify strategies to address constraints for public-sector implementation of G2P approach.

Objective: Monthly teleconference seminars (10 per year) on the theme of "Implementing components of the G2P approach".

Objective: Conduct annual workshop (Manhattan, KS and Fresno, CA) for project participants to provide update on research programs and observe genomic prediction population and managed stress phenotyping.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No, the major goals/specific aims or milestones have not changed. Note, there were some changes in the experimental operations due to the retirement of the UC co-PI and the move of the PI from KSU to CSU (e.g. locations of experiments were changed based on availability of field sites in Colorado vs. California) but the goals have remained the same.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements.



Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

GOAL 1: *Use crop models to characterize the target population of environments (TPE) and predict scenarios under which chilling tolerance and stay green loci provide water-limited yield.*

Objective: *Calibration and model testing for biomass accumulation, flowering time, and commercial sorghum yield under traditional management practices.*

Models used under this goal were the APSIM-sorghum (calibration) and DSSAT-sorghum (calibration and chilling tolerance). Information for model testing regarding crop development and growth resulted from collecting existing information on field experiments, the collaboration between Pioneer and KSU, and the variety testing program led by KSU.

Objective: *Hypothesize trade-offs of early-planted chilling tolerant sorghum and dense-planted stay-green sorghum hybrid-based.*

-Under this goal, we determined the recurrence or dominant water and heat stress patterns (TPE) for sorghum in Kansas, Texas, and Colorado using the APSIM-Sorghum. Similarly, we quantified hypothetical trade-offs of early chilling tolerance and the LT trait. The reason for using the LT trait instead of stay-green is because both traits have similar soil water-saving mechanisms.

GOAL 2: *Complete development of genomic prediction resources, including loci, markers, and germplasm.*

Objective: *Genotype and generate genomic predictions for genome-wide polygenic effects and known major effect loci (height, maturity, chilling tolerance, and stay-green loci).*

Experiments were designed in Kansas and Colorado with two mapping populations (Tx430 x BTx2752 and Tx430 x SC979). Phenotypes collected for these experiments included, canopy temperature, chilling tolerance, stay green, flowering time, and plant height. Genotypic characterization was outsourced via InterteK.



Objective: *Map water-limited yield loci using training population genotyping data and phenotyping data.*

As a proxy for the chilling tolerance, LT, and stay-green, we used seedling vigor, canopy temperature, and chlorophyll content, respectively. To identify genomic variants or QTLs associated with these traits, we used a genome wide association approach. This analysis for the LT trait revealed that two candidate QTLs are likely underlying this trait.

Objective: *Develop KASP markers for novel water-limited yield loci.*

KASP markers were developed for known loci maturity (Ma1, Ma3, Ma6), plant height (Dwarf1–3), stay-green (Stg1,3,5), and chilling tolerance. KASP markers are available to other sorghum breeding & genetics programs in the US via the Intertek-Agritech KASP genotyping platform (<https://www.intertek.com/agriculture/agritech/>). KASP markers are under development for the novel limited-transpiration QTL that were discovered in the preceding objective.

Objective: *Advance genomic prediction population to C1-F7 generation and C2-S2 generation*

Genomic predictions were conducted to determine if flowering time variation in sorghum is governed by epistatic effect. This analysis was performed with the rrBLUP R package using regression models with interaction between genetic factors (Gaussian Kernel) and without interaction (Regression Ridge). Genomic prediction for flowering time and stay green - disregarding the family effect - was 0.77 and 0.5, respectively.

GOAL3: *Collect a third year of managed-stress phenotyping data for the crop model calibration panel and a first year of data for the updated (C1) training population.*

Objective: *Collect data on environment (soil texture, soil water, weather) and crop performance (yield, phenology, biomass) for commercial and experimental hybrids (N = 40) from CA (managed-stress) field sites.*

Experiments were conducted in Kansas in 2019, 2021 in Kansas and in 2022 in Colorado. Fourteen lines, including commercial hybrids were evaluated for crop growth and development. Information collected comprised flowering time, biomass, canopy architecture traits, biomass and yield. These experiments were conducted under well watered in Kansas and managed stress conditions in Colorado



Objective: *Phenotype testcross hybrid updated (C1) training population ($N = 200$) under managed stress (CA) and in sorghum TPE (TX and KS).*

To validate the experimental system, staygreen RILs were evaluated under rainfed conditions in 2021 in a semi-arid environment Eastern Colorado (Akron). Information of flowering time, plant height and stay green score were collected.

The newly developed training populations (Genomics Optimized Recurrent Population; GORP) were phenotyped for agronomic traits (height, flowering time, agronomic suitability) in Kansas and Texas in 2021 and 2022. The GORP for the pollinator parent lines (GORP1R) was phenotyped in Kansas and compared to the sorghum association panel (SAP). The GORP for the seed parent lines (GORP1B) was phenotyped in Texas and compared to the sorghum association panel (SAP). Together, the GORP1R and GORP1B allow genomic mapping in both pollinator and seed parent lines relevant to hybrid grain sorghum breeding.

GOAL 4: *Train early-career researchers (project postdocs and workforce development participants) on technical aspects of G2P approach components and identify strategies to address constraints for public-sector implementation of G2P approach.*

Objective: *Monthly teleconference seminars (10 per year) on the theme of "Implementing components of the G2P approach".*

During the course of the project we had 34 monthly meeting for interdisciplinary learning and exchange on implementation of G2P approaches. Across these meetings, a total eleven early-career scientists presented and received feedback. The early career scientists were from eight institutions (Kansas State, Clemson University, Georgia Southern, University of Nevada, Colorado State, Corteva, USDA) and had backgrounds in 6 plant science disciplines (plant breeding, plant genomics, molecular physiology, crop modeling, agronomy, evolutionary genetics).

Objective: *Conduct annual workshop (Manhattan, KS and Fresno, CA) for project participants to provide update on research programs and observe genomic prediction population and managed stress phenotyping.*

The annual workshop comprised in-person activities in 2019 (KS) and virtual in 2020 and 2021 due to the COVID-19 pandemic. Monthly meetings carried out as planned with participation from up to 12 scientists from up to seven institutions

from 2020 to 2022. Depending on the stage of the project, collaborators presented their research progress, discussed hypotheses and how to test them with experiments, or conducted data analysis to support specific research goals.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

To ensure robust and unbiased results, we have trained the junior scientists performing the research to predefine experimental hypotheses and predictions during the experimental design phase. This approach removes degrees of freedom in the interpretation of findings and prevents inadvertent "p-hacking" (i.e. trying different statistical procedures until a nominally significant p-value is obtained).

- 2.8. List key stakeholders that could be served by results of this project.

Crop breeding programs at public organizations (Texas A&M, University of Nevada-Reno, Purdue University, Clemson University, CIMMYT, USDA).
Crop research programs (Georgia Southern, Colorado State, Kansas State, Kansas State)
Seed companies (Corteva Agriscience, Innovative Seed Solutions, Richardson Seeds, S&W Seeds, Advanta Seeds, etc.)

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

-Community engagement and outreach activities were vital for our research



program success and adoption by the community and stakeholders. To communicate the impact of the G2P research and engage with stakeholders and farmers, scientists participated in several community-based activities, including the governor's conference on the future of water in Kansas (2019 to 2022), presented as speakers for webinar, created outreach materials including newsletters and technical reports, and actively participate in field days.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

COVID restrictions on in-person activities and travel have impeded research in 2020 and 2021. While we have kept our goals/approach the same, we have scaled back the size of a number of laboratory and field activities and switched in-person activities to virtual, where possible.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

- Regular presentations and discussions focused on early-career scientists' professional development (G2P Bridge Accelerate group)

- Training early-career scientists on professional goal-setting and scientific hypothesis-testing

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

- Two postdoctoral scholars at KSU working on genetics and crop modeling, respectively.



- One graduate student (Masters) at KSU working on agronomy and crop physiology.
- Monthly scientific seminars with postdocs, students, early-career faculty, and established faculty.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

- Scholarly publications
- Conference proceedings

- 3.2. Please provide a list of citations for the information products produced during the project.

Raw phenotype and genotype data for Marla et al. 2023 G3 is deposited on Figshare data repository at <https://doi.org/10.6084/m9.figshare.21358191>

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

- a) Carcedo, A.J.P., Mayor, L., Demarco, P., Morris, G.P., Lingenfelser, J., Messina, C.D., Ciampitti, I.A., 2022. Environment Characterization in Sorghum (*Sorghum bicolor* L.) by Modeling Water-Deficit and Heat Patterns in the Great Plains Region, United States. *Front Plant Sci* 13, 768610. <https://doi.org/10.3389/fpls.2022.768610>
- b) Demarco, Paula Andrea, Mayor, L., Messina, C., Prasad, P.V.V., Morris, G.P., Ciampitti, I.A., 2023. Retrospective study in U.S. commercial sorghum breeding: III. Nitrogen internal efficiency. *Crop Science* 63, 879–887. <https://doi.org/10.1002/csc2.20763>
- c) Demarco, Paula A., Mayor, L., Rotundo, J.L., Prasad, P.V.V., Morris, G.P.,

Fernandez, J.A., Tamagno, S., Hammer, G., Messina, C.D., Ciampitti, I.A., 2023. Retrospective study in U.S. commercial sorghum breeding: II. Physiological changes associated to yield gain. *Crop Science* 63, 867–878. <https://doi.org/10.1002/csc2.20845>

- d) Mayor, L., Demarco, P., Lira, S., Fang, Y., Abadie, T., Gambin, B.L., Hammer, G., Ciampitti, I., Cooper, M., Messina, C., 2023. Retrospective study in US commercial sorghum breeding: I. Genetic gain in relation to relative maturity. *Crop Science* 63, 501–510. <https://doi.org/10.1002/csc2.20897>
- e) Raymundo, R., Sexton-Bowser, S., Ciampitti, I.A., Morris, G.P., 2021. Crop modeling defines opportunities and challenges for drought escape, water capture, and yield increase using chilling-tolerant sorghum. *Plant Direct* 5, e349. <https://doi.org/10.1002/pld3.349>
- f) Marla, S., Felderhoff, T., Hayes, C., Perumal, R., Wang, X., Poland, J., Morris, G.P., 2023. Genomics and phenomics enabled prebreeding improved early-season chilling tolerance in Sorghum. *G3 Genes|Genomes|Genetics* 13, jkad116. <https://doi.org/10.1093/g3journal/jkad116>
- g) Raymundo, R., Mclean, G., Sexton-Bowser, S., Morris, G., 2023. Crop modeling suggests limited transpiration would increase yield of sorghum across drought-prone regions of the United States. Accepted with revision at *Frontiers in Plant Science* <https://doi.org/10.1101/2023.06.27.546776>
- h) Raymundo, R., Wang, X., Felderhoff, T., Sexton, S., Poland, J., Lipka, A., Morris, G., 2023. Field phenomics reveals genetic variation for transpiration response to vapor pressure deficit in sorghum. (BioRxiv; Under revision for *Journal Experimental Botany*) <https://doi.org/10.1101/2023.06.23.546345>

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.
- <https://www.ksre.k-state.edu/news/stories/2019/08/unique-public-private-research-maps-the-future-for-dryland-crops.html>
- https://events.cornell.edu/event/FALL_2019_PBG_MORRIS

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project

(U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Yes

Morris, G.P., FELDERHOFF, T., MARLA, S., 2021. Methods and compositions for developing cereal varieties with chilling tolerance. WO2021189034A1.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#1	#0	WO 2021/189034 A1
Copyrights	#0	#0	
Trademarks	#0	#0	
Inventions	#0	#0	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Yes, data which forms the basis for three of our publications (Demarco et al. 2023a Crop Science; Demarco et al. 2023b Crop Science; Mayor et al. 2023 Crop Science) includes proprietary information on plant performance for grain sorghum hybrids from Corteva Agriscience. Summaries of data are publically available in the publications.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

We've described the data sets and findings at scientific conferences, industry visits, and farmer field days in Kansas, Colorado, and Texas.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

- Yes.
- Field phenotyping data for sorghum genotypes under managed drought stress.
- Crop modeling simulation data (yield, flowering time, water use) for sorghum grown under early season chilling and for the LT trait.
- Field phenotyping data for commercial sorghum hybrids to calibrate crop models.
- Raw phenotype and genotype data for chilling tolerance.

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

- All information is related to each other because it corresponds to inbred lines used by public and private programs.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

- Information on crop modeling simulation data for sorghum grown under early season chilling (supplementary file 2).
<https://onlinelibrary.wiley.com/doi/full/10.1002/pld3.349>

- Raw phenotype and genotype data for Marla et al. 2023 G3 is deposited on Figshare data repository at <https://doi.org/10.6084/m9.figshare.21358191>



FOUNDATION FOR
FOOD & AGRICULTURE
RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Geoffrey Morris

PI Signature:  Date: 11/03/2023

Authorized Signing Official (ASO) Name: Catherine Douras

ASO Signature: _____ Date: 11/13/2023